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Effectiveness of generative artificial intelligence-based teaching versus traditional teaching methods in medical education: a meta-analysis of randomized controlled trials

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Abstract

Background Artificial intelligence (AI) has demonstrated remarkable capabilities across diverse medical applications, potentially revolutionizing healthcare delivery systems. This systematic review and meta-analysis investigated the comparative effectiveness of generative artificial intelligence (GAI)-based teaching methodologies versus conventional pedagogical approaches on educational outcomes among medical students.

Methods We conducted a comprehensive literature search across multiple electronic databases, including PubMed, Cochrane Library, EMBASE, and Web of Science, encompassing studies published from January 2014 through January 2025. The review focused on randomized controlled trials (RCTs) that compared GAI-based teaching interventions with traditional instructional teaching methods in medical students.

Results The meta-analysis incorporated 11 eligible RCTs, comprising 786 medical students. Pooled analysis revealed no statistically significant difference in knowledge acquisition scores between GAI-based and traditional teaching approaches (standardized mean difference [SMD] 0.27, 95% confidence interval [CI] -0.31 to 0.85; $p=0.36$). However, subgroup analysis indicated enhanced knowledge performance in the GAI group specifically for extended learning periods (exceeding one week) and practice-oriented courses. GAI-based instruction demonstrated superior outcomes in practical skill development compared to conventional methods (SMD 0.63, 95% CI 0.10–1.16; $p=0.02$). Students in the GAI group reported significantly higher satisfaction scores with their learning experience.

Conclusion While theoretical knowledge acquisition remains comparable between teaching modalities, the distinctive advantages of GAI-based approaches in practical skill development warrant their integration into medical

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curricula. Future research should focus on optimizing the integration of GAI-based teaching methods, standardizing implementation protocols, and evaluating long-term educational outcomes.

Trial registration This protocol was registered on the International Platform of Registered Systematic Review and Meta-analysis Protocols (INPLASY) with the registration number INPLASY202510006. Registered on 2 January 2025.

Keywords Artificial intelligence, Generative artificial intelligence, Medical education

Introduction

Medical education has traditionally relied on lecture-based instruction supplemented with multimedia resources and clinical demonstrations [1]. Recent decades have witnessed evidence-based innovations in medical education. Seminar-based teaching has demonstrated considerable success in enhancing students' active learning capabilities [2], while simulation-based education has shown marked improvements in practical skills and knowledge application [3]. Nevertheless, simulation-based education presents substantial challenges, including significant financial investments in infrastructure and equipment maintenance. Furthermore, these simulations are limited in their ability to comprehensively replicate the full spectrum of clinical scenarios, particularly in emergency and critical care settings [4, 5].

The exponential advancement of artificial intelligence (AI) has transformed various aspects of healthcare delivery, from clinical diagnosis support to patient education [6], necessitating its integration into medical education curricula. A cross-sectional survey spanning 48 countries revealed widespread enthusiasm among medical students regarding AI applications in healthcare, with strong interest in expanding their AI-related educational opportunities [7]. Generative artificial intelligence (GAI), a subset of AI technology based on large language models, creates diverse content through iterative learning from extensive datasets [8, 9]. GAI systems like ChatGPT have demonstrated remarkable capabilities, successfully passing the United States Medical Licensing Examination (USMLE) and achieving performance levels comparable to senior medical students across various licensing exams [10–12]. Research has consistently demonstrated that active learning methodologies substantially enhance educational outcomes [13]. GAI technology offers unique advantages in this context, providing personalized, interactive learning experiences through simulated clinical scenarios and real-time feedback mechanisms. This approach potentially fosters autonomous learning capabilities and sustained educational engagement among students [14–16]. However, the implementation of GAI in medical education has prompted legitimate concerns among educators, particularly regarding potential overreliance on technology and academic integrity issues, including plagiarism and data fabrication [8, 15].

Despite these developments, the empirical evidence supporting the effectiveness of GAI-based teaching methodologies in medical education remains inconclusive. Therefore, this meta-analysis aimed to systematically evaluate the comparative effectiveness of GAI-based teaching approaches versus traditional pedagogical methods in medical education.

Methods

This systematic review and meta-analysis adhered to the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) statement [17]. Ethical approval and patient consent were not required as all analyses were based on previously published studies.

Search strategy and criteria

A comprehensive literature search was conducted by two researchers (L.J. and Y.K.Y.) across multiple electronic databases: PubMed, Cochrane Library, EMBASE, and Web of Science. The search period spanned from January 2014 to January 2025. The analysis included randomized controlled trials (RCTs) that compared GAI-based teaching methodologies with traditional teaching approaches in medical education. Reference lists of eligible studies and relevant review articles were manually screened to identify additional pertinent studies. The detailed search strategy is provided in Appendix 1.

Studies were included if they met all of the following criteria: (1) enrolled medical students as participants; (2) implemented GAI-based teaching methods as the primary intervention; (3) utilized traditional teaching approaches in control groups; (4) reported quantifiable outcomes including knowledge or skill assessment scores; and (5) employed a randomized controlled trial design. Studies were excluded if they were duplicate publications, involved non-medical student populations, lacked genuine GAI tool integration in interventions, did not include a traditional teaching control group, used non-RCT study designs (such as reviews, commentaries, or observational studies), or failed to provide complete and extractable outcome data. Traditional teaching methods are defined as instructor- or expert-led pedagogical approaches where content is delivered primarily through conventional lectures and didactic instruction, without the integration of artificial intelligence technologies in any aspect of the teaching process. GAI-based teaching

methods, in contrast, are characterized by the systematic incorporation of large language model-powered tools (e.g., ChatGPT, conversational AI agents) into the learning environment, where these technologies actively facilitate content delivery, provide personalized feedback, and enhance student-instructor interaction.

Two reviewers (L.J. and Y.K.Y.) independently screened titles and abstracts using EndNote X9 software for reference management. Following initial screening, full-text articles were evaluated for final inclusion. Any disagreements were resolved through consensus discussions with the senior research team (Y.D.W, C.D.X. and J.X.Q.).

Data extraction

Data were extracted as follows: study characteristics (authors, publication date, and country); student characteristics (sample size, grade, and region); GAI measures (tools related to GAI); course features (course type, learning duration, and assessment timing); and outcomes. The primary outcomes were knowledge scores and skill scores after exams. The teaching satisfaction of students was considered as the secondary outcome.

Risk of bias assessment

The methodological quality of included RCTs was evaluated using the Cochrane Collaboration Risk of Bias tool [18]. Six domains were assessed: selection, performance, detection, attrition, reporting, and other biases. Risk levels were categorized as high, low, or unclear. Two reviewers (L.J. and Y.K.Y.) conducted independent assessments, with discrepancies resolved through research team consultation (Y.D.W, C.D.X. and J.X.Q.).

The quality of evidence assessment

Evidence quality was evaluated using the GRADE (Grading of Recommendations Assessment, Development, and Evaluation) framework [19]. Outcomes were classified as very low, low, moderate, or high quality. Disagreements were resolved through the previously described consensus process.

Statistical analysis

Statistical analyses were performed using Review Manager (Version 5.4). Continuous variables were expressed as means with standard deviation (SD). Standardized mean difference (SMD) with 95% confidence interval (CI) was calculated for all continuous variables. The I^2 statistic was employed to assess heterogeneity, with $I^2 > 50\%$ indicating substantial heterogeneity [20]. A random-effects model was employed for outcomes showing substantial heterogeneity ($I^2 > 50\%$) or when meaningful methodological heterogeneity was present (e.g., variations in skill assessment tools and methods), as determined by investigator assessment; otherwise, a fixed-effects model

was used. Predetermined subgroup analyses examined knowledge score variations across four parameters: geographical location (Asia versus Europe), course classification (theory versus practical), learning duration (< 1 week versus ≥ 1 week), and assessment timing (immediate versus delayed). Practice courses were defined as those conducted during clinical rotations or internships, while theoretical courses encompassed all other educational formats. Statistical significance was set at $p < 0.05$.

Results

Literature search

The initial database search yielded 1200 records. Following the removal of 158 duplicates, 1042 titles and abstracts underwent preliminary screening by two independent reviewers. After excluding 1010 ineligible records, 32 full-text articles were assessed for eligibility. Ultimately, 11 studies met the inclusion criteria and were included in the meta-analysis (Fig. 1). A comprehensive list of exclusion reasons is available in the Appendix 3.

Study and participant characteristics

The meta-analysis encompassed 11 RCTs with a total of 786 medical students, comprising 395 participants in GAI-based intervention groups and 391 in traditional teaching control groups [21–31]. All included studies were published between 2024 and 2025, with a predominant representation from Asian institutions (8 studies, 72.7%). The study population primarily consisted of undergraduate medical students, with only one study including postgraduate participants. ChatGPT was the primary GAI tool, utilized in 9 studies (81.8%). 5 studies (45.5%) implemented interventions exceeding one week. 6 studies (54.5%) focused on theoretical courses, while 5 studies (45.5%) addressed practical courses. Detailed baseline characteristics of all included studies are presented in Table 1.

Risk of bias and quality assessment

Regarding random sequence generation, 5 studies were assessed as low risk [21, 22, 24, 26, 29] and one study as high risk [30], with the remaining studies being unclear. For allocation concealment, only one study demonstrated low risk [24], while others were classified as unclear due to insufficient reporting of concealment methods. With respect to blinding, 3 studies reported their blinding methodology [21, 22, 24], though only one study implemented blinding of outcome assessment [23]. Additionally, incomplete outcome data and selective reporting were identified as high-risk domains in 2 studies each [27, 28]. The comprehensive risk of bias assessment for individual studies is visualized in Fig. 2. The overall quality of evidence, as evaluated using the GRADE framework, was categorized as low (Appendix 2). This classification was

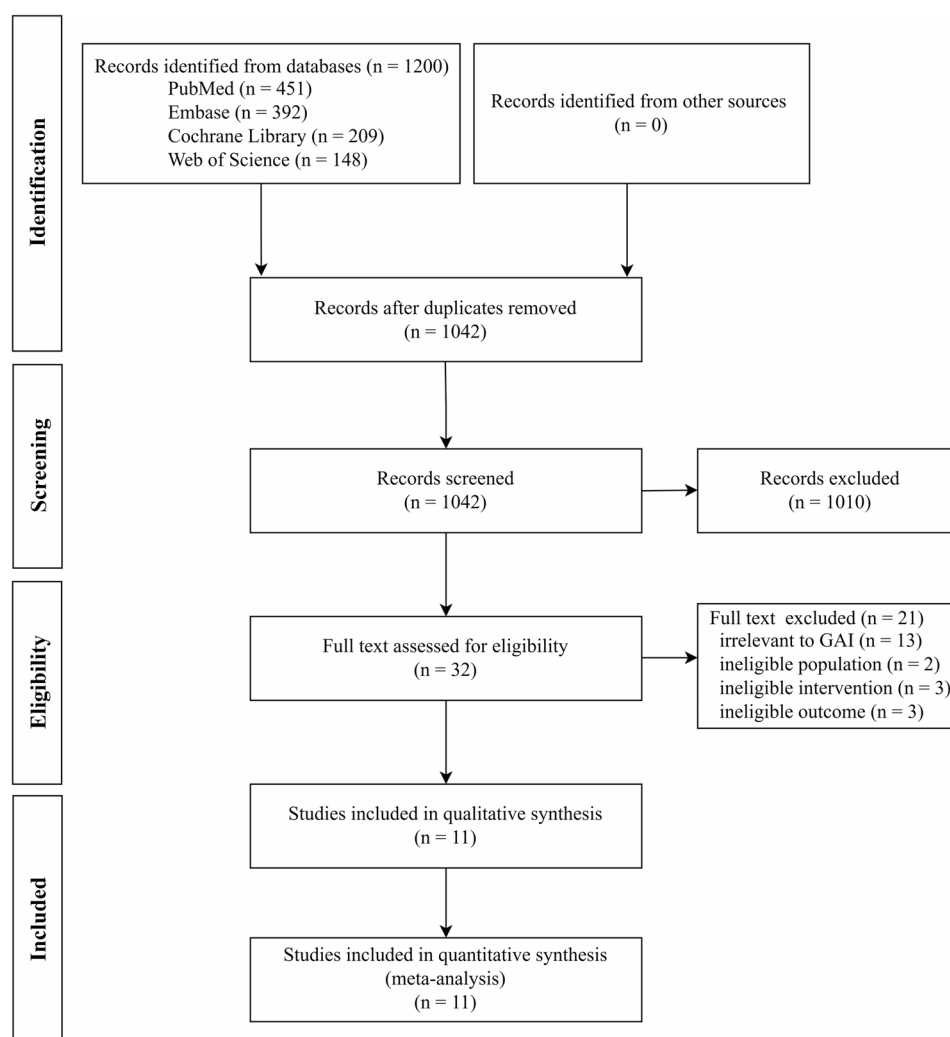


Fig. 1 Flowchart for the literature search and exclusion criteria

primarily attributed to substantial heterogeneity across studies, inadequate reporting of randomization procedures, and limited implementation of blinding methods.

Meta-analysis

Data on knowledge scores were extracted from all eleven studies (395 GAI-based students and 391 traditional teaching students). Meta-analysis with a random-effects model showed that there was no significant difference in knowledge scores between the groups (SMD 0.27, 95% CI -0.31 to 0.85; $p=0.36$; $I^2=93\%$; Fig. 3). Funnel plots showed that no publication bias was observed (Fig. 4). A low quality of evidence was observed as assessed using the GRADE method (Appendix 2). Subgroup analysis revealed that a learning duration of more than one week with GAI could improve knowledge scores ($p=0.02$; Table 2). Moreover, learning with GAI in practice courses resulted in higher knowledge scores ($p<0.01$; Table 2), whereas it had little effect on theory courses. Neither the

geographical location of instruction implementation nor the timing of student achievement assessment influenced the comparative effectiveness of the two teaching methods on theoretical achievement.

Additionally, the meta-analysis, which involved only two studies (52 GAI-based students and 51 traditional teaching students), showed that the GAI-based teaching had significantly higher skill scores compared to the traditional teaching method (SMD 0.63, 95% CI 0.10 to 1.16; $p=0.02$; $I^2=43\%$; Fig. 3), under the random-effects model. There were three studies conducted on the survey on teaching satisfaction in both student groups, and the GAI group reported higher satisfaction than the traditional group (SMD 1.18, 95% CI 0.50 to 1.87; $p<0.01$; $I^2=79\%$; Fig. 3), with the difference being statistically significant. The GRADE quality of evidence is presented in Appendix 2.

Table 1 Baseline characteristics of all included studies

Study	Country	Tradi- tion (n)	GAI (n)	GAI measures	Students	Course type	Learning duration	Assessment timing	Primary Outcomes
Ba et al. 2024 [21]	China	38	39	ChatGPT	Undergraduate	Practice course	2 weeks	Immediate	Knowledge scores and skill assessment
Bhatia et al. 2024 [22]	India	50	50	ChatGPT	Undergraduate	Theory course	30 min	Immediate	Knowledge scores
Çiçek et al. 2024 [23]	Türkiye	65	64	ChatGPT	Undergraduate	Theory course	5 days	Immediate 10 days later	Knowledge scores
Gan et al. 2024 [24]	China	56	54	ChatGPT	Undergraduate	Theory course	1 week	Immediate	Knowledge scores
Huang et al. 2024 [25]	China	32	32	ChatGPT	Undergraduate	Practice course	Not mentioned	Immediate	Knowledge scores
Kavadella et al. 2024 [28]	Europe	38	39	ChatGPT	Undergraduate	Theory course	4 weeks	Immediate	Knowledge scores
Jiang et al. 2024 [27]	China	30	31	The chatbot utilizing standardized patient	Undergraduate	Practice course	2 h	Immediate	Knowledge scores
Svendsen et al. 2024 [29]	Norway	16	15	ChatGPT	Undergraduate Postgraduate	Theory course	45 min	Immediate	Knowledge scores
Tabuchi et al. 2024 [30]	Japan	28	27	AI-generated image	Undergraduate	Theory course	10 min	Immediate	The accuracy rates of test
Wu et al. 2024 [31]	China	30	31	ChatGPT	Undergraduate	Practice course	4 weeks	3 days later	Knowledge and skill scores
Zeng et al. 2025 [26]	China	21	21	ChatGPT	Undergraduate	Practice course	2 weeks	3 days later	Knowledge and skill scores

GAI Generative Artificial Intelligence, ChatGPT Chat Generative Pre-trained Transformer

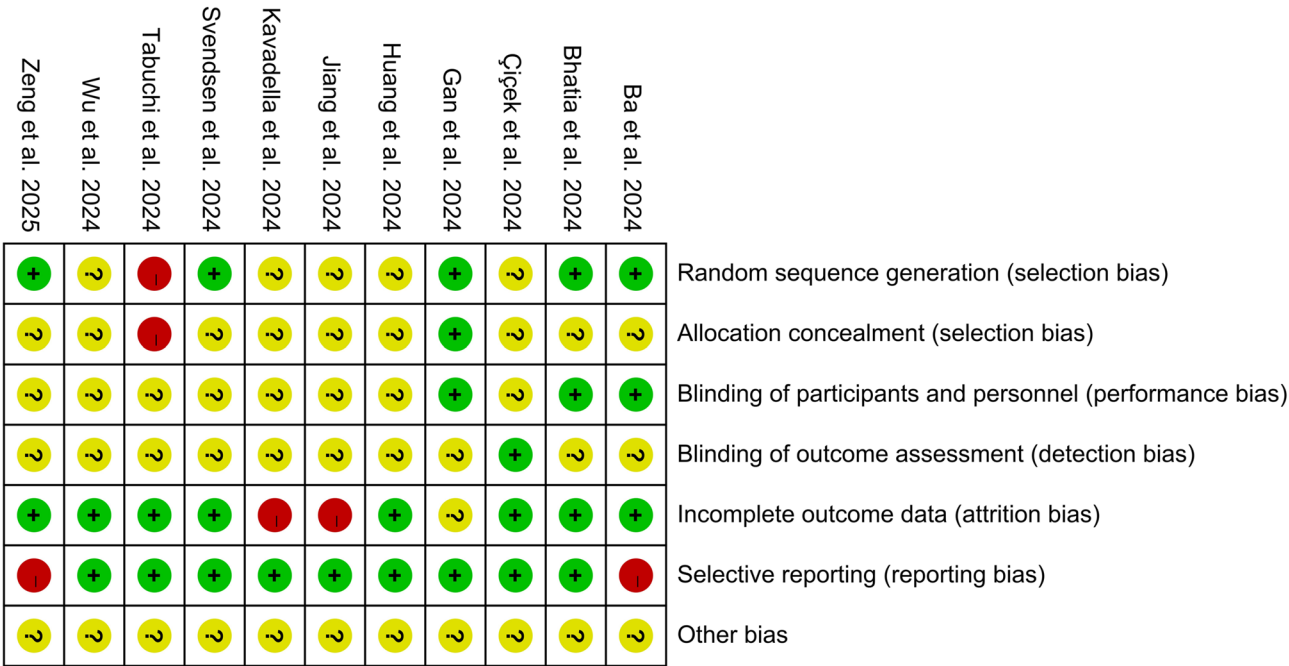


Fig. 2 Risk of bias summary for included studies

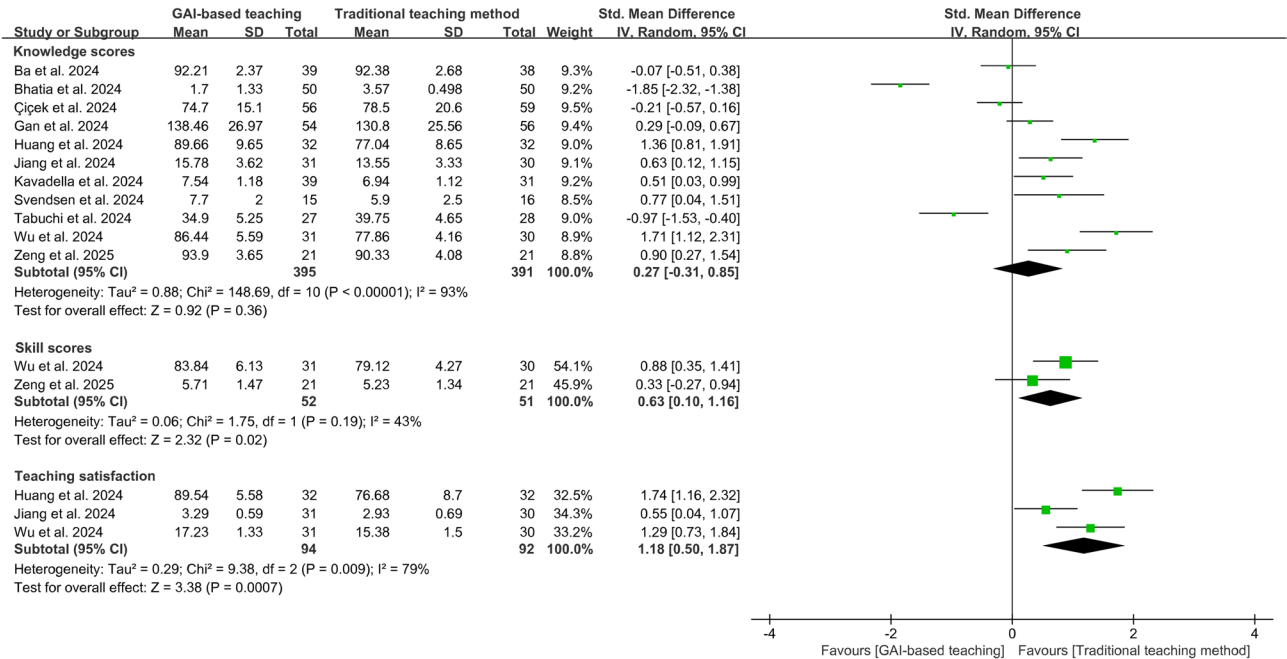


Fig. 3 Forest plot of knowledge scores, skill scores and teaching satisfaction

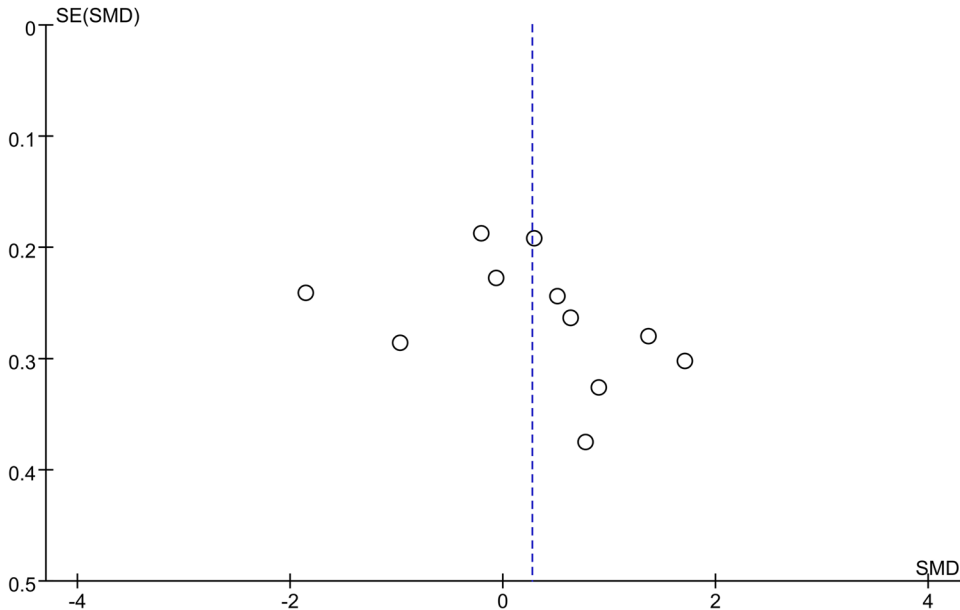


Fig. 4 Funnel plot of knowledge scores

Discussion

Although GAI-based teaching methods have not significantly improved theoretical scores compared with traditional teaching, GAI teaching methods can better improve theoretical scores by extending the teaching time of GAI and adjusting the practical content of GAI courses. As expected, GAI-based teaching methods significantly improve skills and teaching satisfaction.

Although various approaches are employed to evaluate medical students, examination scores remain the

most direct indicator of their knowledge acquisition, skills development, and overall learning outcomes [32]. Our analysis revealed no significant difference in knowledge scores between the two teaching methods; however, this finding should not diminish the potential effectiveness of GAI-based instruction. GAI teaching methodologies, being relatively novel, are still evolving in their pedagogical applications. Notably, our subgroup analysis indicated that GAI demonstrated particular efficacy in practical courses. The limited frequency and duration

Table 2 Subgroup analysis for the knowledge scores

Subgroups	Studies	GAI (n)	Tradition (n)	SMD	95% CI	Model	I ²	P	Subgroup Differences		
									Chi ²	I ²	P
Geographical location											
Asia	8	285	285	0.25	[-0.55, 1.04]	Random	95%	0.55	0.01	0%	0.90
Europe	3	110	106	0.31	[-0.30, 0.92]	Random	77%	0.32			
Learning duration											
< 1 week	5	179	183	-0.34	[-1.26, 0.58]	Random	94%	0.47	3.20	68.8%	0.07
≥ 1 week	5	184	176	0.64	[0.09, 1.19]	Random	84%	0.02			
Course classification											
Theory course	6	241	240	-0.25	[-1.00, 0.50]	Random	93%	0.51	5.19	80.7%	0.02
Practice course	5	154	151	0.89	[0.25, 1.54]	Random	86%	<0.01			
Assessment timing											
Immediate	9	343	340	0.04	[-0.55, 0.64]	Random	93%	0.88	1.39	28.0%	0.24
Delayed	3	108	110	0.82	[-0.32, 1.96]	Random	93%	0.16			

GAI Generative Artificial Intelligence, SMD Standardized Mean Differences, CI Confidence Interval

of GAI implementation in the analyzed studies suggest that extended application periods and enhanced integration may be necessary for comprehensive evaluation. Furthermore, the results may be influenced by confounding factors, including students' established learning patterns [33]. The absence of standardized instructor training protocols and a robust framework for assessing teaching effectiveness presents additional methodological considerations [34]. These identified gaps in current research warrant further investigation and methodological refinement.

In the domain of skill assessment, GAI-based teaching demonstrated superior performance compared to traditional teaching methodologies. Although the analysis of skill scores encompassed only two studies, the significant potential of GAI in medical education warrants attention. Consistent with our subgroup analysis findings, GAI-based methods yielded improved knowledge scores in practical courses relative to theoretical instruction. In contrast to conventional passive knowledge acquisition approaches, GAI facilitates active learning by promoting independent information seeking, inquiry, and problem-solving [35]. This interactive pedagogical approach enhances comprehension while fostering essential cognitive competencies, including critical thinking, problem-solving abilities, and self-directed learning capabilities. GAI addresses traditional constraints of instructor availability and standardized curricula by providing individualized learning experiences and iterative practice opportunities tailored to specific student needs, thereby enhancing learner confidence [36]. These elements are fundamental to the acquisition of complex clinical competencies. Furthermore, GAI effectively simulates diverse clinical scenarios, offering students broader exposure to practical cases and clinical experiences that are typically challenging to replicate through conventional teaching methods [37, 38]. On the other hand, Cook also

highlighted that with thoughtfully designed and appropriately implemented AI-related instructional models, immersive learning experience can be transformed into an accessible, cost-effective, and sustainable educational approach [39]. Consequently, the interactive and adaptive capabilities of GAI should be strategically integrated into guided learning curriculum design to optimize comprehension and knowledge retention.

It is crucial to acknowledge that GAI currently functions primarily as a complementary educational tool rather than a replacement for traditional teaching methodologies in medical education [8, 14, 40]. Without robust curricular integration, GAI's impact on knowledge acquisition and skill development may be constrained, underscoring the necessity for strategic implementation within existing educational frameworks. To address these limitations, researchers have investigated the synergistic potential of AI systems with innovative pedagogical approaches, including flipped classrooms, case-based learning (CBL), and gamified instruction, which may enhance information processing and reinforce foundational knowledge [41–43]. At the same time, the establishment of a practical framework for AI literacy in medical education is imperative for guiding the design, implementation, and evaluation of curriculum content, resource allocation, and intended learning outcomes [44]. Nevertheless, this endeavor places substantial demands on the digital competencies of medical educators. Systematic training for both students and educators is essential to ensure effective GAI utilization and successful adaptation to evolving educational paradigms [45, 46]. Further rigorous research remains necessary to evaluate the longitudinal effects of GAI implementation and establish evidence-based best practices for meaningful integration across diverse curricular contexts.

This meta-analysis exhibits several notable limitations that merit careful consideration. The statistical power of

our findings may be constrained by the restricted number of included studies and their relatively modest sample sizes. Despite conducting extensive subgroup analyses that resulted in reduced heterogeneity, the remaining substantial heterogeneity warrants acknowledgment. A significant methodological concern stems from the quality of included studies, many of which lacked comprehensive documentation regarding randomization procedures and blinding protocols, potentially introducing systematic bias. The limited prior exposure of participants to AI-based educational tools represents an additional confounding factor, as unfamiliarity with these technologies may have influenced learning efficiency and intervention outcomes. Future research should address these limitations through well-designed, large-scale studies employing standardized protocols, while simultaneously investigating the longitudinal impact of GAI integration in educational settings.

Conclusions

This meta-analysis provided evidence regarding the role of GAI-based teaching methodologies in medical education. While our findings indicated comparable theoretical knowledge acquisition between GAI and traditional teaching approaches, GAI demonstrated significant advantages in practical skill development and student engagement. The enhanced outcomes in extended learning periods and practice-oriented courses suggest that GAI's effectiveness is particularly pronounced in specific educational contexts.

However, successful implementation requires careful consideration of several factors: robust curricular integration, adequate technological support, and systematic training for both educators and students. Future research should focus on conducting large-scale, longitudinal studies with standardized methodologies to better understand the long-term impact of GAI integration and establish evidence-based best practices across diverse medical education settings.

Abbreviations

AI	Artificial Intelligence
ChatGPT	Chat Generative Pre-trained Transformer
CI	Confidence Interval
GAI	Generative Artificial Intelligence
GRADE	Grading of Recommendations Assessment, Development, and Evaluation
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses
RCTs	Randomized Controlled Trials
SMD	Standardized Mean Difference

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12909-025-07750-2>.

Supplementary Material 1.

Supplementary Material 2.

Supplementary Material 3.

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None.

Authors' contributions

L.J. and Y.K.Y. provided study design, literature screening, and data extraction. All authors contributed to manuscript writing, review, and editing and approved the final submitted version.

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Data availability

The datasets and analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

Ethical approval and patient consent were not required as all analyses were based on previously published studies.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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